

Team Name _____

SECTION 1: Aquatic Ecology & Water Systems (25 points)

Multiple Choice (1 point each, 10 points total)

- | | |
|---------|----------|
| 1. _B__ | 6. _B__ |
| 2. _C__ | 7. _C__ |
| 3. _D__ | 8. _A__ |
| 4. _C__ | 9. _C__ |
| 5. _B__ | 10. _B__ |

Fill in the Blank (1 point each, 10 points total)

- 11. littoral
- 12. secondary
- 13. usable
- 14. invasive
- 15. eutrophication
- 16. coagulation
- 17. bioaccumulation
- 18. potable
- 19. decomposers
- 20. sedimentation

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Short Answer

21. Sedimentation impacts (3 points - 1.5 point each, check off any two): level of detail= tiebreaker 1

- ☐ Smothers eggs/organisms on bottom
- ☐ Blocks sunlight/reduces photosynthesis
- ☐ Clogs fish gills
- ☐ Fills in habitats/reduces depth
- ☐ Reduces water clarity/increases turbidity

22. "decrease" or "decreases" (1 point)

- Explanation: Warm water holds less oxygen than cold water (1 point) extra: gas particles move faster when they get warm and are able to escape the attraction to water particles (tiebreaker 2)
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SECTION 2: Macroinvertebrate & Aquatic Species Identification (25 points)

Part A: Pollution Tolerance Classification (1 points each, 5 points total)

23. Stonefly: Class __1__
24. Dragonfly: Class __2__
25. Blood midge: Class __4__
26. Water strider: Class __5__
27. Dobsonfly: Class __1__

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Part B: Water Quality Indicators (8 points total)

28. B

29. C

30. Purple loosestrife impacts (2 points - 1 point each, check off any two):

- ☐ Outcompetes/displaces native plants for space, water, nutrients, or sunlight
- ☐ Reduces food sources for wildlife (low nutritional value compared to natives)
- ☐ Eliminates/degrades nesting and feeding habitat for native animals (especially waterfowl and insects)
- ☐ Reduces plant biodiversity/species diversity in wetlands
- ☐ Alters wetland water flow or hydrology (changes how water moves through wetland)
- ☐ Blocks access to open water for wildlife

31. Zebra mussel impacts (2 points - 1 point each, check off any two):

- ☐ Filters excessive amounts of water removing food/phytoplankton for native species (starves other filter feeders)
- ☐ Outcompetes native mussels for food and space
- ☐ Clogs water intake pipes and infrastructure (economic and functional damage)
- ☐ Alters water clarity dramatically (can disrupt ecosystem balance and allow more light penetration)
- ☐ Attaches to and smothers native mussels (directly kills them)
- ☐ Sharp shells cut and harm wildlife, fish, and humans

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Identification (2 points each, 12 points total)

A water boatman	B water penny	C mayfly
D damselfly	E crane fly	F Dobsonfly

SECTION 3: Water Quality Data & Testing Parameter Knowledge (2 point each, 30 points total)

32. B

37. D

42. B

33. C

38. B

43. B

34. C

39. B

44. A

35. D

40. A

45. B

36. B

41. B

46. A